Proposals for Using Auctions for Airport Landing Slots

INFO6220: Networks II: Market Design

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1 Description of the Problem

Imagine a bustling airport like John F. Kennedy International Airport (JFK), where planes circle overhead, waiting for a chance to land, while passengers grow frustrated with delays. Airport congestion occurs when the demand for landing and takeoff slots outstrips the limited capacity of runways, creating a bottleneck that ripples across the aviation network. The article from Yahoo News reports that Seattle-Tacoma International Airport (Sea-Tac) experienced over 700 flight delays on May 16, 2025, due to a global technology outage affecting multiple industries. The outage, linked to a faulty software update from CrowdStrike, impacted Microsoft systems, causing widespread disruptions. Nationally, over 29,000 flights were delayed and nearly 7,000 canceled, with ripple effects continuing into the weekend [1].

The challenge is to allocate these scarce slots efficiently, ensuring airlines can operate smoothly while keeping airports safe and passengers happy. Key stakeholders include airlines, airports, air traffic control authorities, and passengers. Airlines chase profitable schedules, coveting peak-time slots to maximize ticket sales. Airports strive to balance revenue, safety, and throughput, often caught between competing demands. Air traffic authorities focus on system-wide coordination to prevent gridlock, and passengers simply want on-time flights at reasonable prices.

The trouble is, these incentives clash: airlines may hold slots to block competitors, even if they don't fully use them, while airports struggle to reallocate, leading to overcrowded schedules and delays [2]. This misalignment demands a smarter allocation system to distribute slots efficiently.

2 Explanation of the Market Inefficiency

The root of the problem lies in the outdated system of slot allocation, often called "grandfathering" under International Air Transport Association (IATA) guidelines, where airlines keep slots based on historical usage [3]. Picture an airline holding onto a prime morning slot at a busy airport, using it for a half-empty flight just to keep competitors out. This system fails to match slots with current demand, leading to underused runways, blocked new entrants, and a market that's anything but efficient. Economists would call this a Pareto-inefficient outcome—some airlines could get better slots without harming others, yet the system prevents it [4]. New airlines, especially low-cost carriers, face high barriers to entry, stifling competition and innovation. Congestion worsens as slots remain locked with incumbents, even when others value them more [5]. A centralized market mechanism, like an auction, could fix this by allocating slots based on airlines' true willingness to pay, unlocking efficiency and opening the skies to fairer competition. A combinatorial auction, using a Vickrey-Clarke-Groves (VCG) design, addresses the complexity of airport slot allocation [4]

is essential to match slots with current demand, enhance competition, and alleviate congestion.

3 Description of the Mechanism

Enter the combinatorial auction, a sophisticated tool designed to tackle the complexity of airport slot allocation [4]. Unlike first-price or English auctions, which risk speculative bidding, or simple Vickrey auctions, which treat slots individually, combinatorial auctions allow airlines to bid on slot bundles (e.g., \$12,000 for {08:00 landing, 08:30 takeoff}) to create cohesive schedules [6]. Why does this matter? Airlines don't just want one slot; they need a set that works together, like puzzle pieces forming a flight plan. A Vickrey auction, while strategy-proof and straightforward, might leave an airline with a landing slot but no matching takeoff, rendering it useless [6]. In a combinatorial auction, airlines submit bids for their desired bundles, attaching a dollar value (e.g., \$10,000 for {Slot 08:00, 08:30}). The airport authority, acting as the auctioneer, runs an optimization algorithm to assign bundles in a way that maximizes total value while respecting constraints, like runway capacity, which is shown in Fig. 1. By using a Vickrey-Clarke-Groves (VCG) design, the auction can be strategy-proof, meaning airlines bid their true valuations without gaming the system, ensuring slots go to those who value them most [4]. This mechanism transforms chaos into order, aligning schedules with demand and easing congestion.

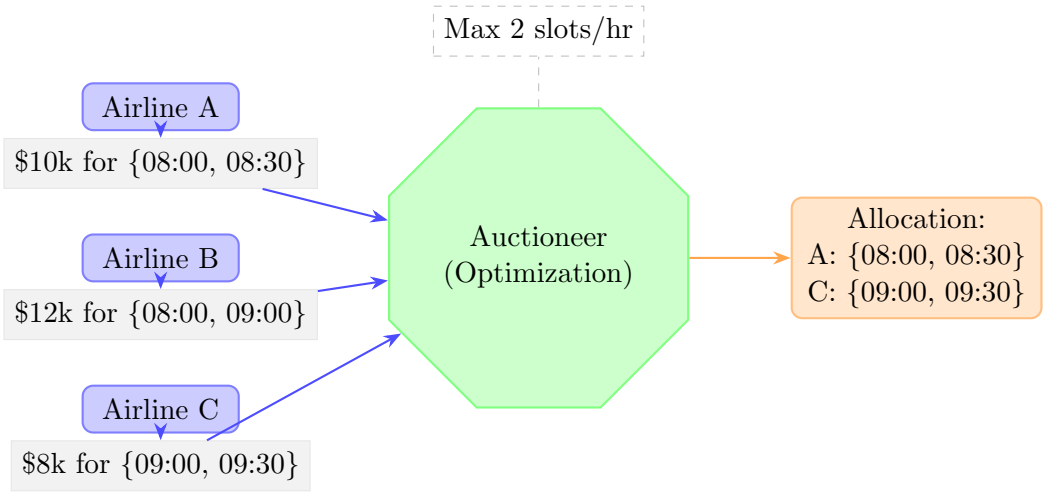


Figure 1: Combinatorial auction process for airport slots. The diagram illustrates the process of a combinatorial auction for allocating airport landing slots, as described in the report. It visually represents the interaction between airlines (bidders), the auctioneer (e.g., airport authority), and the optimization process that determines slot allocations.

4 Critical Evaluation

The combinatorial auction shines in several ways. It delivers allocative efficiency by matching slots to airlines with the highest valuations, making every runway minute count and cutting

delays [4]. Its VCG design discourages sneaky bidding, fostering trust in the process. Plus, it levels the playing field, giving smaller airlines a shot at prime slots, which could spark competition and lower fares [5]. Imagine a new budget airline landing at a major hub, offering affordable flights because it won a fair bid.

However, the auction is not perfect. The mechanism's computational complexity is a headache—solving the winner determination problem for a busy airport like JFK requires serious computing power. Its computational complexity demands robust systems, challenging for busy airports [6]. Smaller airlines, strapped for cash, might struggle to compete with deep-pocketed giants, raising ethical questions about access equity. Practically, many airlines may lobby against auctions, fearing loss of their slot monopolies, and regulators must guard against collusion to keep the process transparent. Compared to alternatives like congestion pricing, which charges higher fees for peak slots, auctions offer more flexibility but demand more setup [2]. Table 1 captures these trade-offs. Subsidies or tiered bidding could mitigate financial barriers for smaller airlines.

Table 1: Evaluation of Combinatorial Auction for Airport Slots

Advantages	**Disadvantages**
1. Maximizes efficiency by valuing bundles	1. Computationally complex, requiring robust systems
2. Strategy-proof with VCG, ensuring honest bids	2. Financial strain on smaller airlines
3. Opens slots to smaller airlines, boosting competition	3. Risks of regulatory pushback and collusion

5 Conclusion

The combinatorial auction offers a promising fix for the tangled mess of airport slot allocation, replacing outdated grandfathering with a system that rewards value and efficiency. It aligns with course concepts like Pareto efficiency and strategy-proofness while tackling real-world congestion. Yet, its complexity and potential to sideline smaller players call for careful design—perhaps blending auctions with subsidies for new entrants or simpler pricing models. By addressing these challenges, airports could become gateways to smoother skies and fairer competition.

References

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Appendix

This report leverages LLM model (Grok) to generate the diagram in Latex, with the given prompt: "Please help me draw the combinatorial auction process for airport slots in latex by the given description: Airlines (Left Side): Three nodes labeled "Airline A," "Airline B," and "Airline C" are depicted as blue rounded rectangles with a light blue fill, positioned vertically at coordinates (0,2), (0,0), and (0,-2). These represent bidders competing for slots, reflecting the competitive bidding emphasized in Le et al. (2003). Bids (Middle Left): Each airline connects to a bid node via a thick blue arrow pointing right. The bid nodes, shown as light gray rectangles, are located at (3,2), (3,0), and (3,-2). They display specific bundle bids: "$10k for 08:00, 08:30" (Airline A), "$12k for 08:00, 09:00" (Airline B), and "$8k for 09:00, 09:30" (Airline C), illustrating bundle bidding from Ball et al. (2018). Auctioneer (Center): A green octagonal node at (6,0), resembling a gear, is labeled "Auctioneer (VCG Optimization)." The gear shape symbolizes the computational complexity of the optimization algorithm, as noted in Stern & Turvey (2003), and the VCG label highlights the strategy-proof mechanism from Ball et al. (2018). Blue arrows from each bid node converge to this auctioneer, indicating bid submission. Capacity Constraint (Above Auctioneer): A dashed gray rectangle at (6,3) labeled "Max 2 slots/hr" is connected to the auctioneer by a dashed gray line. This represents runway capacity limits, a key congestion factor from Levine (1969). Allocation (Right Side): An orange rounded rectangle at (10,0) labeled "Allocation: A: 08:00, 08:30, C: 09:00, 09:30" shows the output of the auction, connected to the auctioneer by a thick orange arrow. This reflects the efficient slot assignment to high-value bidders, as described in Ball et al. (2018). Visual Style: The diagram uses a clean, color-coded design (blue for airlines/bids, green for auctioneer, orange for allocation, gray for constraints) with small font for readability. Arrows are thick and use the Stealth tip for clarity." And then adjust the parameters for the Figure 1.